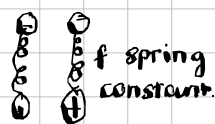


Oscillator Model

To understand material's response, it is useful to understand oscillator's model.

Assume the net charge system is 0 and applied by an external field E



When there is a spring, we call Lorentz

$$m\ddot{x} = -fx - \frac{m}{\tau}\dot{x} - qE \Rightarrow \ddot{x} + \frac{1}{\tau}\dot{x} + \omega_0^2 x = -\frac{qE}{m} \quad \omega_0^2 = \frac{f}{m} \quad 2\gamma = \frac{1}{\tau}$$



When the electron is free, we call Drude model $f=0$

$$m\ddot{x} = -\frac{m}{\tau}\dot{x} - qE \Rightarrow \ddot{x} + \frac{1}{\tau}\dot{x} = -\frac{qE}{m}$$

The force by f is due to the binding interaction between electrons and protons. The friction γ is due to factors such as electrons scattering, interactions with holes, etc.

Using either FT, one arrives:

$$d(\omega) = -qX(\omega) = -\frac{q^2 E(\omega)}{m} \frac{1}{\omega^2 - \omega_0^2 + i\omega\gamma}$$

With $P(\omega) = X(\omega) E(\omega)$, we have the response.

$$X(\omega) = -\frac{Ne^2}{m} \frac{1}{\omega^2 - \omega_0^2 + i\omega\gamma} = -\frac{Ne^2}{m} \frac{1}{2\omega_0\gamma} \left[\frac{1}{\omega - \omega_0 + i\gamma} - \frac{1}{\omega + \omega_0 + i\gamma} \right] \quad \omega_0^2 = \omega_0'^2 - \gamma^2$$

This form contains so much information:

Let $\Delta = \omega - \omega_0'$, we see immediately poles

$$\Delta + i\gamma = 0 \Rightarrow \omega = \omega_0' - i\gamma$$

The single number contains two pieces of physics

$$\text{Re}(\omega) = \omega_0' \quad \text{Im}(\omega) = -\gamma \Rightarrow e^{-i(\omega_0' - i\gamma)t} = e^{-i\omega_0' t} e^{-\gamma t}$$

an oscillation at ω_0' with a decay of γ

The poles are unachievable. Realistically, we scan " ω " as the real axis for response while γ deviates from the real axis

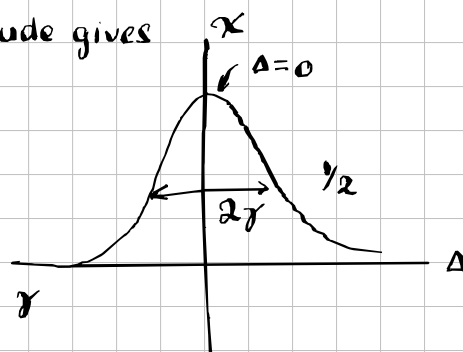
Now, look at $X(\omega) = |X(\omega)| e^{i\phi(\omega)}$, the amplitude gives

$$|X|^2 \propto \frac{1}{\Delta^2 + \gamma^2} \quad \text{Lorentzian}$$

$$\text{At } \Delta=0, |X|^2 \propto \frac{1}{\gamma^2} \Rightarrow \text{maximum}$$

$$\text{At half maximum, } \frac{1}{\Delta^2 + \gamma^2} = \frac{1}{2} \frac{1}{\gamma^2} \Rightarrow |\Delta| = \gamma$$

$$\text{The full width at half maximum FWHM} = 2\gamma = \frac{1}{\tau}$$



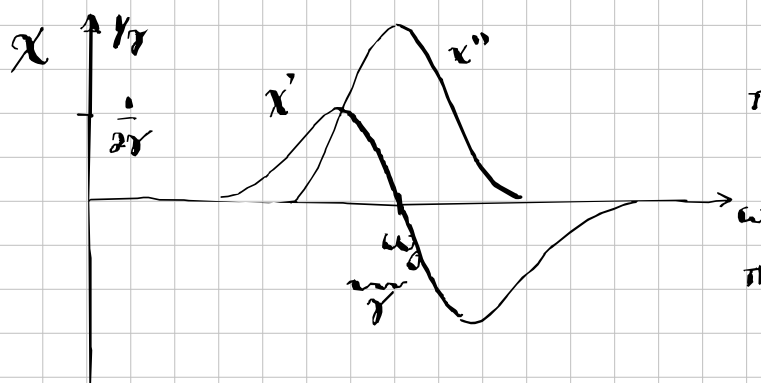
The phase is induced such that the created polarization is out of phase. Leading to a bad interference of induced E field and incoming E field.

$$\phi(\omega) = -\arctan\left(\frac{\gamma}{\Delta}\right) = \arctan\left(\frac{X''}{X'}\right)$$

At $\Delta=0$, the shift $\phi(\omega) = -\pi/2 \Rightarrow$ exactly the cause of delay.

The same form can be seen in a different form

$$\frac{1}{\Delta + i\gamma} = \frac{X'}{\Delta^2 + \gamma^2} - \frac{X''}{\Delta^2 + \gamma^2}$$



We call X' as dispersion and

X'' as absorption, but one needs care.

It is wrong to say X' - phase only and X'' - amplitude.

Take $E(t) = E_0 \cos(\omega t)$

$$P(\omega) = X(\omega) E(\omega)$$

$$= E_0 [X' \cos \omega + X'' \sin \omega]$$

The X' is in phase with $E(t)$, X'' is out of phase

X' describes how materials absorb and

relieve energy. In one cycle, no net gain.

But, it "rescales" $E_0 \rightarrow E_0 + X'$ amplitude.

The term X'' removes energy from the field. $E \frac{dP}{dt}$

$$E \frac{dP}{dt} = E_0^2 \omega [-X' \cos^2 \omega + X'' \sin^2 \omega]$$

These are real reasons not sth as $e^{iX'} e^{-X''}$

* When $\omega \approx \omega_0$, a large absorption occurs